

Verification using UVM : Reporting Macro

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- UVM Print Macros
 - `uvm_info`
 - `Uvm_warning`
 - `Uvm_error`
 - `Uvm_fatal`

UVM Print Macros

In UVM , \$display is dangerous

- Always prints
- Cannot be filtered
- Floods logs
- Breaks regressions

UVM gives print macros

- uvm_info
- uvm_warning
- uvm_error
- uvm_fatal

UVM macros connect to reporting systems which understands

- Severity
- Verbosity
- Component hierarchy

UVM Print Macros

``uvm_info (ID, MESSAGE, VERBOSITY)`

Example

- ``uvm_info("DRV", "Driving Transaction, UVM_MEDIUM)`
- ``uvm_warning("CFG", "Config not found")`
- ``uvm_error ("SB", "Data Mismatch")`
- ``uvm_fatal("CFG", "Virtual Interface missing")`
- Only `uvm_info` is controlled by verbosity
- Warning, error, and fatal always print

Verbosity Levels

``uvm_info (ID, MESSAGE, VERBOSITY)`

Example

- ``uvm_info("DRV", "Driving Transaction, UVM_MEDIUM)`



Verbosity	Description
UVM_NONE	Always printed, verbosity configuration cannot disable it
UVM_LOW	Prints a message if verbosity is configured as UVM_LOW or above
UVM_MEDIUM	Prints a message if verbosity is configured as UVM_MEDIUM or above
UVM_HIGH	Prints a message if verbosity is configured as UVM_HIGH or above
UVM_FULL	Prints a message if verbosity is configured as UVM_FULL or above
UVM_DEBUG	Prints a message if verbosity is configured as UVM_DEBUG

Example : Verbosity Levels

```
1  `include "uvm_macros.svh"
2  import uvm_pkg::*;
3
4  module tb;
5  int data;
6  initial begin
7  #50;
8  `uvm_info("TB_TOP","Hello World",UVM_LOW);
9  $display ("Hello World with display");
10 `uvm_info("TB",$sformatf("Value of data: "%0d)",data),UVM_NONE);
11 end
12 endmodule
```

Example : Verbosity Levels

```
1  `include "uvm_macros.svh"
2  import uvm_pkg::*;
3
4  module tb;
5  initial
6  begin
7    uvm_top.set_report_verbosity_level(UVM_LOW);
8    $display("Default verbosity level : %0d", uvm_top.get_report_verbosity_level);
9    `uvm_info("TB_TOP","Hello world with UVM_DEBUG",UVM_DEBUG);
10   #10;
11   `uvm_info("TB_TOP","Hello world with UVM_FULL",UVM_FULL);
12   #10;
13   `uvm_info("TB_TOP","Hello world with UVM_HIGH",UVM_HIGH);
14   #10;
15   `uvm_info("TB_TOP","Hello world with UVM_MEDIUM",UVM_MEDIUM);
16   #10;
17   `uvm_info("TB_TOP","Hello world with UVM_LOW",UVM_LOW);
18 end
19 endmodule
```

Verbosity Levels

- UVM Verbosity levels are numeric thresholds
- A message prints only if its verbosity value is less than or equal to the current simulation verbosity
- Example
 - **Simulation command** : `+UVM_VERBOSITY=UVM_HIGH`
 - **Inside code** : `uvm_top.set_report_verbosity_level(UVM_HIGH)`

Component level

- `drv.set_report_verbosity_level(UVM_HIGH);`
- `mon.set_report_verbosity_level(UVM_LOW);`

Default Actions

- **UVM_INFO:** Trigger if Verbosity is less than the default Verbosity threshold
- **UVM_WARNING:** Cannot be filtered out
- **UVM_ERROR:** Store the trigger count in UVM_COUNT
- **UVM_FATAL :** Call \$finish and stop simulation

https://github.com/edaplayground/edaplayground/blob/master/docs/_static/uvm-1.2/src/base/uvm_report_object.svh

Verbosity Levels

Verbosity Level	Value
UVM_NONE	0
UVM_LOW	100
UVM_MEDIUM	200
UVM_HIGH	300
UVM_FULL	400
UVM_DEBUG	500

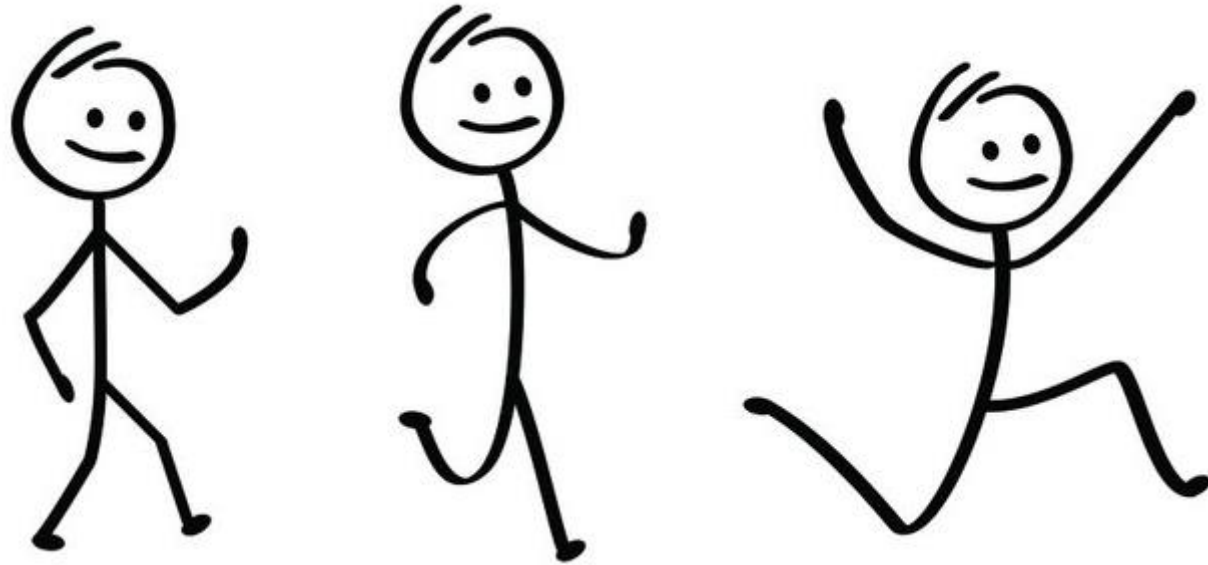
- Debug mode (HIGH verbosity)
- If the default verbosity set to
UVM_MEDIUM : 200

Message	Value	Compare with 200	Printed ???
UVM_LOW	100	100 <= 200	YES
UVM_MEDIUM	200	200 <= 200	YES
UVM_HIGH	300	300 <= 200	NO
UVM_FULL	400	400 <= 200	NO

Other Print macros

```
1 `include "uvm_macros.svh"
2 import uvm_pkg::*;
3
4 class driver;
5 task run();
6     `uvm_info("DRV","Information Mesage",UVM_NONE);
7     `uvm_warning("DRV","Potential Error");
8     `uvm_error("DRV","Real Error");
9     #10;
10    `uvm_fatal("DRV","Simulation cannot continue");
11    $display("Simulation done");
12 endtask
13 endclass
14
15 module tb;
16 driver drv;
17 initial begin
18     drv=new();
19     drv.run();
20 end
21 endmodule
```

KERNEL: --- UVM Report Summary ---
KERNEL:
KERNEL: ** Report counts by severity
KERNEL: UVM_INFO : 2
KERNEL: UVM_WARNING : 1
KERNEL: UVM_ERROR : 1
KERNEL: UVM_FATAL : 1
KERNEL: ** Report counts by id
KERNEL: [DRV] 4



Thank you !

Happy Learning